

REVIEW

Current and Novel Biomarkers for Diagnosis and Subtyping of Primary Aldosteronism

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ABSTRACT: Primary aldosteronism is a common and potentially curable cause of secondary hypertension, yet it remains substantially underdiagnosed. Current diagnostic pathways rely on the aldosterone-to-renin ratio, confirmatory tests, and subtype classification by imaging and adrenal venous sampling. However, these approaches are limited by assay variability, medication effects, imperfect diagnostic accuracy, and the absence of a true reference standard, particularly for bilateral disease. Recent molecular imaging and histopathologic data further challenge the traditional binary distinction between unilateral and bilateral primary aldosteronism and instead support a biological continuum of renin-independent aldosteronism. In this context, novel biomarkers may improve both detection and biological subtyping. Steroidomics is currently the most clinically advanced approach, with plasma and urinary steroid profiles showing promise for primary aldosteronism detection, identification of *KCNJ5*-mutated adenomas, and subtype prediction. Proteomics and circulating microRNAs provide complementary information on systemic effects, upstream regulation, and lesion biology, while multiomics strategies increasingly demonstrate that integrated models outperform single-modality approaches. Nevertheless, translation into clinical practice remains constrained by small and selected cohorts, heterogeneous analytical platforms, inconsistent reference standards, and limited external validation. Future progress will depend on standardized assays, interpretable machine learning models, and prospective multicenter implementation studies linking biomarker-guided diagnosis to improved clinical outcomes.

Key Words: adenoma ■ biomarkers ■ hypertension ■ machine learning ■ proteomics

Primary aldosteronism (PA) is a leading and potentially reversible cause of secondary hypertension, characterized by autonomous aldosterone secretion independent of renin-angiotensin system regulation. This endocrine disorder promotes sodium retention, potassium excretion, and sustained hypertension, but its clinical impact extends beyond blood pressure elevation. Excess aldosterone exerts direct pathogenic effects on the cardiovascular system, contributing to myocardial fibrosis, vascular dysfunction, and renal injury, thereby conferring a higher risk of adverse cardiovascular outcomes compared with primary hypertension.^{1–4} Despite a prevalence estimated at 5% to 10% among hypertensive individuals^{5–8} and substantially higher in resistant hypertension and hypokalemia (up to 45%), PA remains underdiagnosed due to limitations in current screening and diagnostic workflows.

The diagnostic evaluation of PA involves sequential biochemical screening, confirmatory testing, and subtype classification to distinguish unilateral from bilateral

disease and establish treatment strategy. Adrenalectomy offers the potential for cure in unilateral PA, whereas bilateral disease requires long-term pharmacological blockade of mineralocorticoid receptors.^{9,10} While recent guidelines advocate broader screening strategies,^{11,12} this has increased the demand for accurate and scalable diagnostic tools. Thus, improving diagnostic precision remains a key unmet need in the management of hypertensive populations.

CURRENT DIAGNOSTIC PATHWAY

The contemporary diagnostic pathway for PA has evolved from a rigid stepwise model to a more flexible, likelihood-based approach, although the supporting evidence remains limited and low in quality.^{11,12} The recommendation for screening extends to all patients with hypertension with initial evaluation reliant on measurements of aldosterone and renin. A positive test is defined by suppressed renin in combination with

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Nonstandard Abbreviations and Acronyms

AVS	adrenal venous sampling
EV	extracellular vesicle
LC-MS/MS	liquid chromatography with tandem mass spectrometry
miRNA	microRNA
ML	machine learning
PA	primary aldosteronism
PV	pathogenic variant

inappropriately elevated aldosterone and an increased aldosterone-to-renin ratio (ARR), interpreted within the clinical context and potential confounders (eg, medications, potassium status). Renin-dependent and sex-specific aldosterone cutoffs increase the specificity of the ARR from 44% to 71% in women and from 73% to 82% in men.¹³ Following a positive screening result, subsequent steps are guided by the likelihood of PA and therapeutic intent (Figure 1). In patients with a high probability of PA (eg, resistant hypertension, hypokalemia, markedly suppressed renin with elevated aldosterone), confirmatory tests may be omitted, and diagnosis can be confirmed directly. In patients with intermediate

probability, confirmatory tests (eg, the seated saline suppression test) are suggested,^{14,15} particularly if there is consideration of surgical management. In patients who are not candidates for or do not desire surgery, empirical treatment with a mineralocorticoid receptor antagonist may be initiated based on pathological screening results alone. For patients with a low probability of PA (normokalemia and plasma aldosterone concentrations below a specific threshold), mineralocorticoid receptor antagonist treatment is suggested based on a pathological ARR due to low or suppressed-renin levels.¹²

Following disease confirmation, and for candidates for surgical treatment, subtype differentiation is required to guide therapeutic decision-making (Figure 1). Anatomic imaging with CT and MRI is widely used for subtype evaluation in PA, but its diagnostic accuracy is limited.^{16,17} These modalities cannot assess hormonal activity, risking misclassification due to nonfunctional incidentalomas or undetected aldosterone-producing microadenomas. Data show only moderate sensitivity (64%) and low specificity (58%) of anatomic imaging for lateralization of PA.¹⁸ Thus, adrenal venous sampling (AVS) remains the standard approach for subtype classification. AVS provides direct assessment of lateralized aldosterone secretion and is therefore critical to identify patients for adrenalectomy.¹²

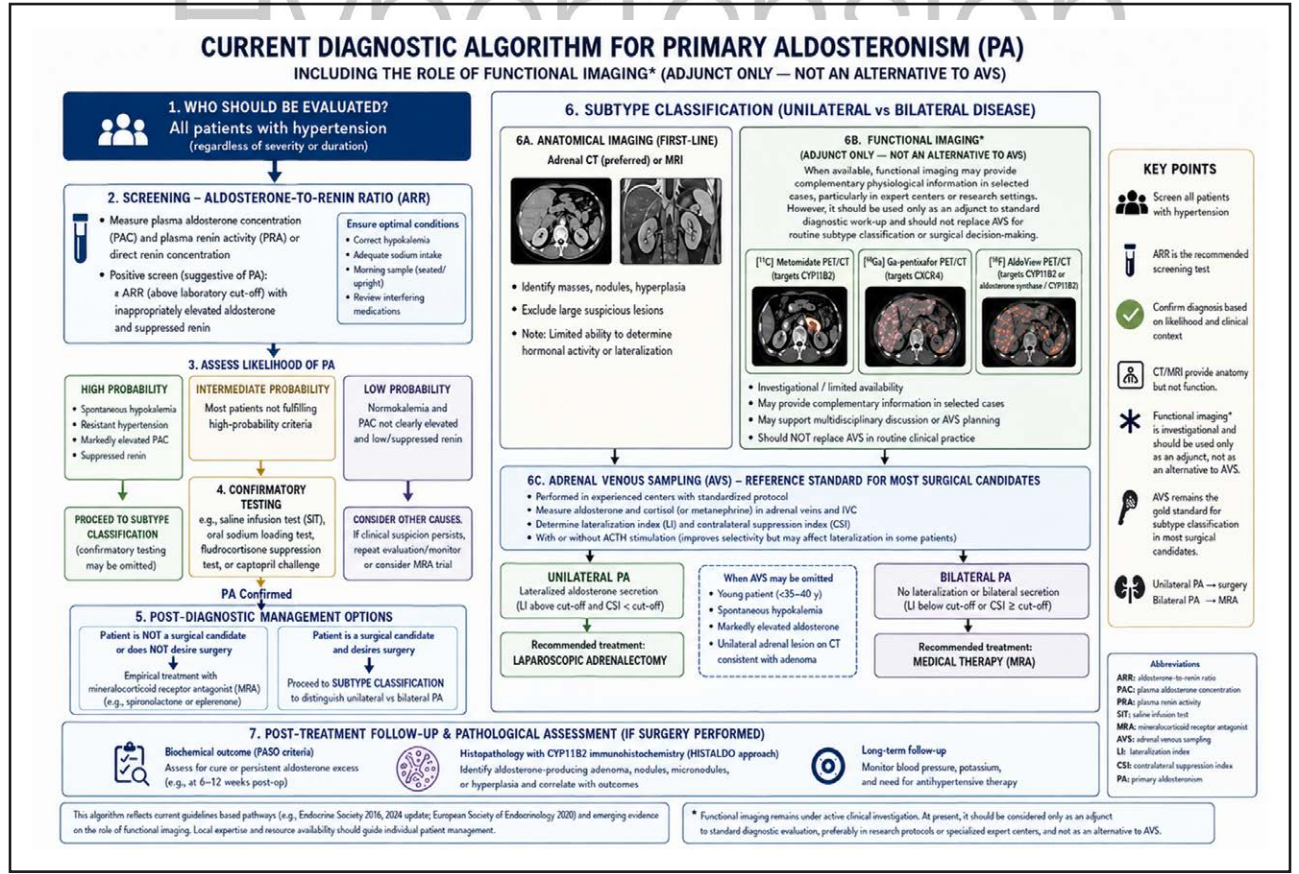


Figure 1. Current diagnostic work flow for primary aldosteronism.

Recent evidence, though, challenges the long-standing position of AVS as the gold standard for subtyping PA.¹⁹ Indeed, interpretation of AVS results has limitations. In the study by El Ghorayeb et al,²⁰ ACTH stimulation improved bilateral AVS selectivity from 66.7% to 91.8%, but basal and post-ACTH lateralization differed in 28% of cases, most often because lateralized cases at baseline were reclassified as bilateral after stimulation. These discrepancies may reflect protocol-related variability, biological heterogeneity, and variable ACTH responsiveness. Apparent bilateral aldosterone suppression can further complicate AVS interpretation, usually as an analytical artifact, but also through sampling from nonrepresentative adrenal tributaries or transient quiescence of aldosterone secretion.²¹ Importantly, discordance between AVS and functional imaging raises further uncertainty about the utility of AVS to distinguish unilateral from bilateral disease. In particular, in the prospective MATCH study,^{22,23} [¹¹C]metomidate PET-CT was evaluated in 143 patients with PA, of whom 128 had follow-up and 78 underwent adrenalectomy. The study demonstrated comparable accuracy to AVS in predicting postadrenalectomy outcomes, with no superiority of either modality across Primary Aldosteronism Surgery Outcome end points.²⁴ Indeed, [¹¹C]metomidate PET-CT proved noninferior, but not superior, to AVS for predicting biochemical and clinical success after adrenalectomy (72.7% and 65.4% versus 63.6% and 61.5%, respectively). Notably, concordance between AVS and [¹¹C]metomidate PET-CT for unilateral disease was low ($\approx 30\%$) due to a substantial proportion of cases (37%) classified as unilateral by only 1 modality. This resulted in an unusually high prevalence of unilateral PA ($\approx 67\%$), largely driven by additional cases detected by [¹¹C]metomidate PET-CT. Despite this discordance, surgical outcomes were similar regardless of the diagnostic modality used, raising fundamental concerns about the reliability of AVS.

Other molecular imaging approaches have also shown promise. In the prospective CONPASS study, [⁶⁸Ga]Ga-pentixafor PET/CT showed 90% concordance with AVS in 100 patients, increasing to 100% in those with a unilateral adrenal nodule ≥ 10 mm.²⁵ Performance was lower in micronodules or atypical glands, and surgical confirmation was incomplete. A subsequent 160-patient study reported that combining 18-oxocortisol with [⁶⁸Ga]Ga-pentixafor PET/CT achieved 95.2% sensitivity and correctly classified 87.2% of cases, although blinded assessment was not explicitly stated.²⁶ [¹⁸F]AldoView, designed as a highly selective PET tracer for aldosterone synthase, represents a particularly promising emerging molecular imaging approach for PA.²⁷ In an initial clinical evaluation including 3 healthy volunteers and 15 patients with adrenal lesions, [¹⁸F]AldoView was well tolerated and detected positive lesions in surgically confirmed aldosterone synthase-positive APA or nodules, whereas no positive lesions were observed in idiopathic

hyperaldosteronism, Cushing syndrome, or nonfunctioning adrenal adenomas.²⁸ This tracer may offer several advantages over other molecular imaging modalities. Unlike [⁶⁸Ga]Ga-pentixafor, which targets CXCR4 and therefore serves as a surrogate marker of aldosterone-producing lesions, [¹⁸F]AldoView directly targets aldosterone synthase. This may be particularly relevant in small or heterogeneous lesions, where CXCR4 expression, nodule size, and interpretation criteria may limit [⁶⁸Ga]Ga-pentixafor performance. Compared with [¹¹C]metomidate, [¹⁸F]AldoView was developed to improve selectivity for aldosterone synthase (CYP11B2) over 11 β -hydroxylase (CYP11B1), potentially reducing reliance on dexamethasone pretreatment to suppress 11 β -hydroxylase-related adrenal background uptake. However, current evidence remains preliminary, based on small and selected cohorts, and requires prospective validation in larger PA populations against AVS, CYP11B2-based histopathology, and postoperative biochemical and clinical outcomes before routine clinical implementation.

HISTOPATHOLOGY FOR DIAGNOSIS AND SUBTYPING OF PA



Histopathology is increasingly reshaping the conceptual framework of PA by challenging the traditional dichotomy of unilateral versus bilateral disease. A key methodological advance underlying this shift was the development of monoclonal antibodies against human aldosterone synthase to enable reliable immunohistochemical identification of abnormal aldosterone-producing tissue and characterization of the cellular pathology of PA.²⁹ The term lateralized PA may be more appropriate than unilateral PA, as some patients may show asymmetrical bilateral aldosterone secretion rather than truly one-sided disease.³⁰ Accordingly, although AVS remains the main preoperative tool for localization, postoperative biochemical success according to Primary Aldosteronism Surgery Outcome criteria is the most clinically meaningful reference standard in surgically treated patients (Figure 1). Persistent aldosterone excess after adrenalectomy may reflect either preexisting asymmetrical bilateral disease or an incorrectly identified diseased adrenal.²⁴ In this context, the HISTALDO framework, integrating aldosterone synthase immunohistochemistry with conventional morphology, distinguishes classical lesions, typically a solitary APA or dominant nodules, from nonclassical patterns such as multiple aldosterone-producing nodules, micronodules, or diffuse hyperplasia.^{29,31} This distinction appears clinically relevant. Meyer et al³² analyzed 95 adrenal specimens from AVS-classified unilateral PA and found more hyperplasia in patients with absent or partial biochemical success than in those with complete success (49% versus 21%; $P=0.004$), and fewer solitary functional adenomas (44% versus 79%; $P<0.001$). In a subgroup of 70 patients with available steroid profiling,

lack of successful biochemical outcome or bilateral PA was predicted with 76% accuracy by discriminant analysis and 86% by decision-tree analysis. These findings were supported by a prospective single-center cohort of 60 operated patients, in which complete biochemical success was achieved in 97.6% of classical versus 66.7% of nonclassical cases.³³ In addition, nonclassical cases showed a higher contralateral adrenal vein-to-peripheral vein aldosterone ratio on AVS, indicating no suppression of aldosterone production from the unressected adrenal gland and a possible explanation for the lower biochemical cure rate. Thus, available evidence supports a spectrum model of PA, in which apparent unilateral lateralization may, in some patients, reflect markedly asymmetrical bilateral aldosterone production rather than strictly unilateral disease.³⁴

RATIONALE FOR NOVEL BIOMARKERS

Like many disorders, aldosterone dysregulation appears to exist along a biological continuum, from subclinical or mild abnormalities to overt biochemical and clinical disease. Nevertheless, clinical practice requires actionable diagnostic criteria. Current diagnostic tests, are typically derived from outcome-based or consensus-driven frameworks. Such thresholds balance sensitivity and specificity and help identify patients likely to benefit from further investigation or intervention. Without such

decision boundaries, it would be unclear when to initiate confirmatory testing, imaging, AVS, or treatment, increasing the risk of both under- and overdiagnosis. A key unresolved challenge is, therefore, to develop preferably noninvasive biomarkers and probabilistic diagnostic tools that better reflect the continuous nature of PA, capture disease probability, severity, and subtype heterogeneity, and preserve clinically implementable decision thresholds. Such approaches could also detect subclinical aldosterone excess among patients currently classified as having low-renin hypertension, although its frequency and clinical significance remain uncertain.

In this direction, novel biomarkers, such as proteomics, microRNA (miRNA) profiling, metabolomics, and broader multiomics approaches, examine different but complementary layers of disease biology and, therefore, offer distinct opportunities for biomarker discovery in PA (Figure 2). Proteomics focuses on proteins, the functional products of gene expression, and can reveal alterations in enzyme abundance, signaling pathways, and tissue-derived phenotypes. In contrast, miRNA profiling assesses small noncoding RNAs that regulate gene expression at the posttranscriptional level and may, therefore, capture earlier upstream regulatory changes. Metabolomics provides a dynamic snapshot of biochemical activity by measuring small molecules and metabolic intermediates, including steroid hormones and their precursors (steroidomics). Measurements are typically by mass spectrometry-based methods. These approaches

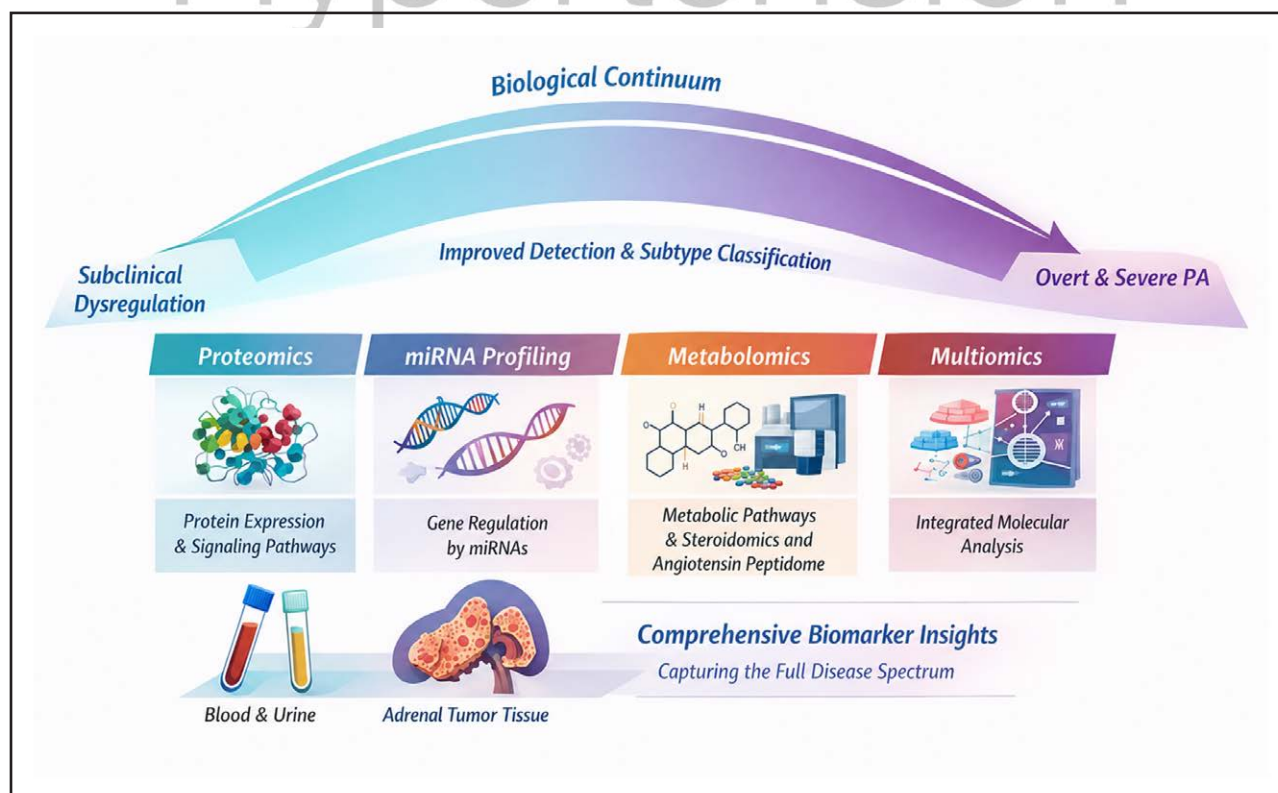


Figure 2. Biological and analytical foundations of novel biomarkers in primary aldosteronism (PA).

can be applied both to biospecimens, such as blood and urine, and to adrenal lesions, where they may better reflect local disease mechanisms. Multiomics integrates ≥ 2 of these data layers, often together with clinical or imaging information, to generate a more comprehensive view of disease biology and may therefore be especially useful for capturing the biological continuum of PA and improving diagnosis, subtype classification, and treatment stratification. To better appreciate the potential role of novel biomarkers in the diagnosis and subtyping of PA, it is essential to understand the pathways of aldosterone synthesis, the molecular mechanisms involved in the pathophysiology of PA, and the analytical techniques available for the assessment of the biomarkers.

ALDOSTERONE BIOSYNTHESIS PATHWAY

Aldosterone is synthesized in the zona glomerulosa of the adrenal cortex through a tightly regulated enzymatic cascade. Under physiological conditions, aldosterone production is controlled primarily by angiotensin II and extracellular potassium, with ACTH exerting a more transient modulatory effect. These stimuli regulate acute cholesterol mobilization and, chronically, transcription of *CYP11B2*, the gene encoding aldosterone synthase, largely through calcium- and cAMP-dependent signaling pathways.³⁵ Aldosterone synthase is a mitochondrial enzyme normally restricted to zona glomerulosa cells and catalyzes the final sequential conversions of deoxycorticosterone to corticosterone, 18-hydroxycorticosterone, and ultimately aldosterone. Mineralocorticoid-specific steroidogenesis is maintained by adrenal zonation, with *CYP11B2* expression restricted to zona glomerulosa cells and the absence of *CYP17A1*, thereby preventing cortisol precursor synthesis in this compartment. This prevents the generation of 17 α -hydroxylated intermediates, such as 11-deoxycortisol, and thereby limits shunting toward glucocorticoid and androgen pathways, which are characteristically located in the zona fasciculata and zona reticularis, respectively.³⁶ In PA, the normally stimulus-dependent and zonally restricted regulation of adrenal steroidogenesis is disrupted, leading to autonomous *CYP11B2* transcription, increased aldosterone synthase expression, and excessive aldosterone production. This disturbed zonation may also create the biochemical conditions required for hybrid steroid formation. In support of this concept, Guo et al³⁷ reported that aldosterone-producing adenomas that secrete excessive amounts of hybrid steroids were usually composed of zona fasciculata-like rather than zona glomerulosa-like cells, suggesting that hybrid steroid excess may reflect aberrant steroidogenic cell identity as well as dysregulated enzyme expression. In particular, 18-oxocortisol synthesis appears to depend on the convergence of *CYP11B2* activity with the availability of 17 α -hydroxylated substrates, especially 11-deoxycortisol.³⁸ Recent evidence suggests that this may occur through coexpression

of *CYP17A1* and *CYP11B2* in a small subset of zona glomerulosa cells, thereby allowing intracellular generation of 11-deoxycortisol and its subsequent conversion by aldosterone synthase to 18-oxocortisol.³⁹ Alternatively, 11-deoxycortisol may diffuse from cells expressing *CYP17A1*, and potentially downstream steroidogenic enzymes such as 21-hydroxylase (*CYP21A2*), interspersed among aldosterone synthase (*CYP11B2*)-positive glomerulosa cells, where it may serve as a paracrine substrate for 18-oxocortisol synthesis.^{37,40} These findings support the concept that disorganized adrenal steroidogenesis, rather than simple coexpression of aldosterone synthase (*CYP11B2*) with 17 α -hydroxylase/17,20-lyase (*CYP17A1*), provides the mechanistic basis for increased production of hybrid steroids such as 18-hydroxycortisol and 18-oxocortisol in PA.^{36–41}

MOLECULAR DRIVERS OF PA

At the molecular level, PA is driven predominantly by somatic pathogenic variants (PVs) in genes regulating ion channels and intracellular calcium homeostasis. These PVs converge on a common pathogenic mechanism: increased intracellular calcium signaling, which stimulates *CYP11B2* expression and aldosterone overproduction. The most frequently identified PVs occur in the *KCNJ5* gene, which encodes a G protein-activated inward rectifier potassium channel.³⁰ PVs disrupt potassium selectivity, leading to sodium influx, membrane depolarization, and activation of voltage-gated calcium channels.⁴² This results in sustained calcium entry into adrenal cells, driving aldosterone synthesis. Additional PVs include those in *CACNA1D*, *ATP1A1*, and *ATP2B3*, all of which similarly affect ion transport and calcium signaling.^{43–45} *CACNA1D* PVs lead to gain of function of voltage-dependent calcium channels, directly increasing calcium influx. In contrast, *ATP1A1* and *ATP2B3* PVs impair ion exchange mechanisms, disrupting sodium-potassium and calcium homeostasis, respectively, and indirectly promoting membrane depolarization and calcium accumulation.^{43,44} Despite differing molecular origins, these PVs ultimately converge on a shared pathway of calcium-mediated activation of aldosterone synthase (Figure 3). Additional genes with PVs have also been identified, expanding the molecular landscape of PA beyond the canonical driver genes. These include rarer alterations in *CACNA1H*, *CLCN2*, *MCOLN3*, *SLC30A1*, *CTNNB1*, *GNA11/GNAQ*, and *CADM1*, highlighting additional mechanisms involving dysregulated ion handling, Wnt/ β -catenin signaling, and impaired intercellular communication.⁴⁶

The aforementioned genetic alterations are associated with distinct steroidogenic phenotypes.⁴⁷ For example, *KCNJ5*-mutated tumors typically exhibit marked overproduction of hybrid steroids, whereas *KCNJ5* wild-type tumors may display more subtle biochemical

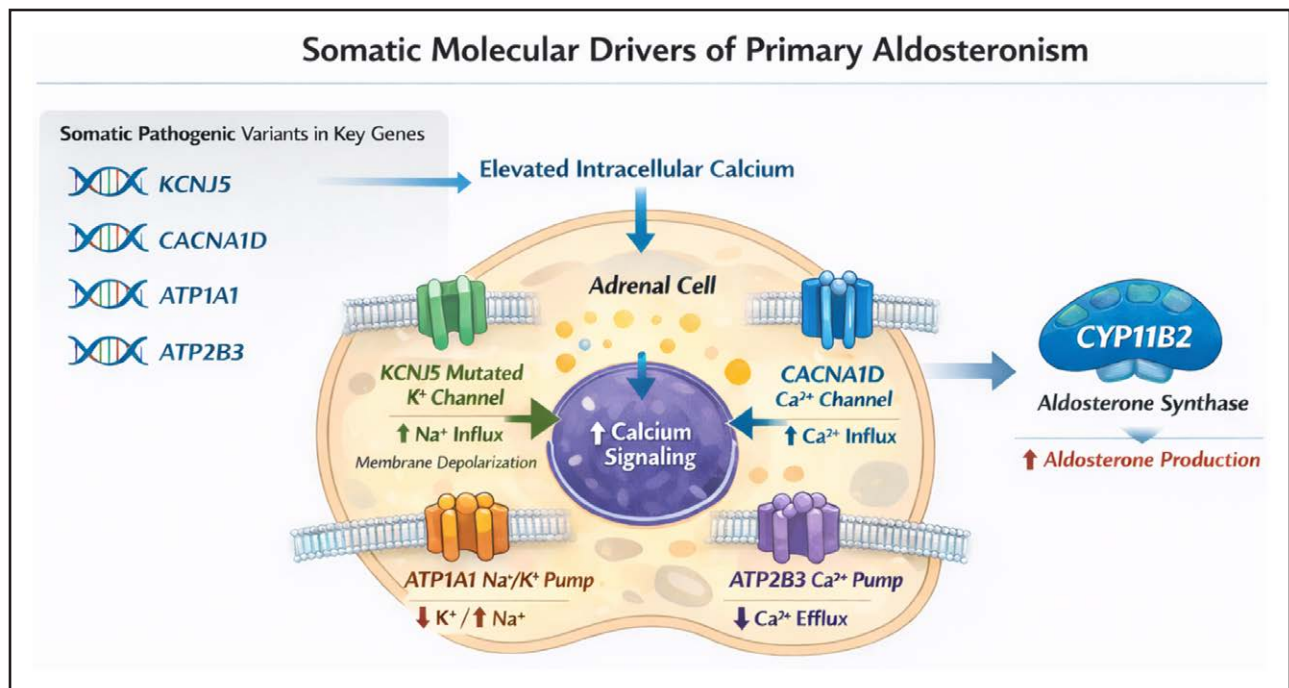


Figure 3. Key genes harboring pathogenic variants in primary aldosteronism.



profiles. In particular, *KCNJ5*-mutated adenomas are often associated with hypokalemia, higher aldosterone levels, larger adenoma size, and a distinct steroidogenic profile characterized by increased production of hybrid steroids.⁴⁸ Further phenotypic characteristics of *KCNJ5*-mutated APA include a fasciculata-like cellular composition and lack of aldosterone responsiveness to upright posture, consistent with altered adrenal zonation and dysregulated control of aldosterone secretion. They are disproportionately more frequently found in younger individuals, and women.^{49,50} These genotype-phenotype correlations support the concept that molecular heterogeneity in PA is directly mirrored in circulating steroid profiles, thus providing a mechanistic foundation for biomarker development.

ANALYTICAL TECHNIQUES

Two principal analytical approaches are currently used for the measurement of biomarkers: immunoassays and mass spectrometric methods. Immunoassays remain widely used in routine clinical laboratories because they are accessible, automated, high-throughput, and relatively inexpensive. They differ by assay format and detection system. With regard to format, competitive immunoassays, commonly used for small molecules such as steroids, rely on competition between endogenous and labeled analytes for limited antibody-binding sites. Noncompetitive, or sandwich, immunoassays use 2 antibodies recognizing distinct epitopes and are, therefore, better suited to larger molecules such as peptides and proteins. With regard to detection, radioimmunoassays

are based on radiolabeled tracers and often include extraction or purification steps, whereas modern chemiluminescence immunoassays use luminescent detection and usually measure aldosterone directly in serum or plasma. Because steroids such as aldosterone are low-molecular-weight haptens, most aldosterone immunoassays remain competitive, which contributes to susceptibility to analytical interference from matrix effects, protein-binding effects, heterophile antibodies, and platform-specific bias.^{51–54} In contrast, mass spectrometric methods, although not totally free from analytical interferences,⁵⁵ provide higher analytical specificity through sample extraction, chromatographic separation, parent-ion selection, fragmentation, and multiple-reaction monitoring, allowing more accurate and multiplexed quantification of low-molecular-weight analytes.

The aforementioned analytical differences are clinically relevant for the diagnostic work-up of PA, as ARR measurements show substantial interlaboratory variability and inconsistent diagnostic thresholds.¹² In the prospective within-patient study by Fuld et al,⁵⁶ including 604 evaluable patients, liquid chromatography with tandem mass spectrometry (LC-MS/MS)-derived and immunoassay-derived ARR showed no clinically relevant difference in screening performance. In particular, at 95% sensitivity, both methods achieved only 57% specificity, underscoring the limited precision of ARR-based case detection. In contrast, assay choice appears more consequential for the SIT. In a 5-center study of 240 patients, Liaison and iSYS immunoassays measured aldosterone concentrations 86% and 58% higher than LC-MS/MS, respectively, resulting in 62 out of 162

immunoassay-positive tests that were nonpathological by LC-MS/MS.¹⁴

Analytical platforms for novel biomarkers vary according to the biological layer being interrogated and can be applied to both circulating biospecimens and adrenal tissue. Proteomics is typically assessed by targeted or discovery LC-MS/MS workflows in plasma or tissue.⁵⁷ miRNA profiling is usually performed by small-RNA sequencing followed by qPCR validation.⁵⁸ Metabolomics and steroidomics may be targeted or untargeted and are now commonly performed using LC-MS/MS in plasma, urine, or tissue, whereas GC-MS was previously the reference approach for urinary steroid profiling but has been progressively replaced in many settings by LC-MS/MS-based methods.^{59,60} Overall, these platforms capture complementary aspects of PA biology and may support biologically informed diagnosis and subtyping through novel biomarkers.⁶¹

METABOLOMICS

Steroid metabolomics has emerged as one of the most biologically informative strategies in PA (Figure 2). Dysregulated steroidogenesis generates characteristic metabolite fingerprints, including hybrid steroids such as 18-oxocortisol and 18-hydroxycortisol, that reflect aberrant adrenal enzymatic activity and may associate with specific molecular or clinical subtypes.^{47,59} Indeed, elevated levels of 18-oxocortisol and 18-hydroxycortisol are strongly associated with unilateral PA, especially those harboring *KCNJ5* PVs. These metabolites can reach markedly higher concentrations in unilateral disease compared with bilateral hyperplasia or essential hypertension, making them valuable markers for both diagnosis and subtyping. In addition to hybrid steroids, intermediates of the mineralocorticoid pathway, such as corticosterone and 11-deoxycorticosterone derivatives, provide insight into upstream enzymatic activity. Alterations in these metabolites can contribute to distinguishing PA from other hypertensive states. However, their precise role in the pathogenesis of hypertension remains unclear.⁶²

Plasma Steroids

Plasma steroid profiling by LC-MS/MS has emerged as a promising noninvasive adjunct for the diagnosis of PA (Table). In a first study to explore this technology, machine learning (ML) models based on 7 plasma steroids identified PA with 69% sensitivity and 94% specificity, and the addition of steroid profiling to the ARR increased the proportion of correctly identified PA cases from 69.6% to 81.3% at similar specificity, indicating that multisteroid panels provide clinically relevant information beyond conventional screening.⁵⁹ A further practical advantage is that steroid-based screening appears less vulnerable

than the ARR to interference from antihypertensive treatment. Indeed, in the prospective multicenter cohort of 839 tested patients reported by Constantinescu et al,⁶³ interfering medications significantly impaired the diagnostic performance of the ARR, whereas steroid-probability scores were minimally affected overall.

With regard to subtype classification, early evidence from a multicenter study of 216 patients with PA, showed that combinations of peripheral plasma steroids correctly classified 80% of unilateral PA versus bilateral disease.⁶⁴ Indeed, 18-oxocortisol concentrations were 8.5-fold higher in unilateral PA than in bilateral disease, supporting hybrid steroids as informative markers of unilateral disease (Table). This biological signal appears particularly strong in *KCNJ5*-mutated unilateral PA: in a study of 79 patients with genotyped unilateral PA, peripheral and adrenal venous steroid profiles showed 16-fold and 18-fold higher 18-oxocortisol concentrations respectively, in *KCNJ5*-mutated tumors than in other genotypes, and a 7-steroid peripheral fingerprint correctly classified 92% of unilateral PA by genotype.⁴⁷ The strongest clinical data to date for PA classification come from the multicenter study that involved 273 patients and identified unilateral PA with pathogenic *KCNJ5* variants with 85% sensitivity and 97% specificity; similar performance was observed in the validation cohort.⁵⁹ However, these results also clarify the current limitation of plasma steroid profiles for subtype work-up. The performance is best for biologically distinctive lesions, particularly *KCNJ5*-mutated and larger unilateral PA, but weaker for the full spectrum of unilateral disease. This was illustrated by Yang et al,⁶⁵ who studied 197 patients (33 micro-unilateral PA, 95 macro-unilateral PA, 69 bilateral PA) and found that plasma steroid profiles classified macro-unilateral and bilateral PA with 93% specificity but had poor specificity for micro-unilateral PA (33%); combining steroid profiling with CT and selective AVS improved micro-unilateral PA classification to 68% and overall classification to 92%. Thus, plasma steroid profiling has clear value for enriching the probability of unilateral, particularly *KCNJ5*-associated, disease and for improving biological subtyping. In carefully selected cases, particularly when CT shows a unilateral adrenal mass and the steroid profile is strongly suggestive of a *KCNJ5* PV, steroid profiling may even serve as an alternative to AVS. Nevertheless, current evidence is still limited by referral-center cohorts, lack of prospective studies, AVS-based reference standards, lower performance in small *KCNJ5* wild-type lesions, and the need for broader assay standardization before this approach can be adopted routinely.

Steroid profiling has also been proposed as an adjunct to AVS, particularly to improve selectivity and interpretation (Table). Conventional AVS relies on cortisol gradients to confirm successful cannulation, but cortisol has an important limitation: its long circulatory

Table. Representative Studies of Emerging Biomarkers for Diagnosis and Subtyping of Primary Aldosteronism

Biomarker class	Study	Cohort/design	Matrix/platform	Main clinical task	Main findings/performance	Main limitations
Plasma steroidomics	Eisenhofer et al, JAMA Netw Open 2020 ⁵⁹	Multicenter; 632 patients tested for PA	Plasma; LC-MS/MS	Diagnose PA; identify KCNJ5-associated unilateral disease	7-Steroid ML model identified PA with 69% sensitivity and 94% specificity; adding steroid profiling to ARR increased correctly identified PA cases from 69.6% to 81.3% at similar specificity; identified unilateral APA with pathogenic <i>KCNJ5</i> variants with 85% sensitivity and 97% specificity	Referral-center cohorts; SIT and AVS-based reference standard; best performance in biologically distinctive lesions
Plasma steroidomics	Constantinescu et al, Hypertension 2024 ⁶³	Prospective multicenter cohort; 839 tested patients	Plasma steroid-probability scores	Robustness under antihypertensive treatment	Interfering medications significantly impaired ARR performance, whereas steroid-probability scores were minimally affected	Does not fully solve subtype classification; depends on assay standardization
Plasma steroidomics	Eisenhofer et al, Clin Chem 2016 ⁶⁴	216 patients with PA	Peripheral plasma steroids	Distinguish APA from BAH	Steroid combinations correctly classified 80% of APA vs BAH; 18-oxocortisol 8.5-fold higher in APA than BAH	Moderate cohort size; subtype reference dependent on existing clinical standards
Plasma steroidomics/genotype	Williams et al, Hypertension 2016 ⁴⁷	79 patients with genotyped unilateral PA	Peripheral and adrenal venous steroids	Identify genotype-specific disease	18-Oxocortisol concentrations were 16-fold higher in peripheral blood and 18-fold higher in adrenal venous blood in <i>KCNJ5</i> -mutated tumors; a 7-steroid peripheral fingerprint correctly classified 92% of APA by genotype	Focused on unilateral, genotyped disease; less generalizable to full PA spectrum
Plasma steroidomics and imaging	Yang et al, J Steroid Biochem Mol Biol 2019 ⁶⁵	197 patients: 33 micro-APA, 95 macro-APA, 69 BAH	Plasma steroids plus CT/selective AVS	Subtype classification, especially micro-APA	Steroid profiles classified macro-APA vs BAH with 93% specificity, but specificity for micro-APA was only 33%; combining steroid profiling with CT and selective AVS improved micro-APA classification to 68% and overall classification to 92%	Reduced performance in small lesions; shows steroids alone are insufficient for identification all unilateral cases
AVS adjunct steroids	Alessi et al, Clin Endocrinol (Oxf) 2026 ⁷¹	229 patients with PA	Adrenal/peripheral venous steroids by LC-MS/MS	Improve AVS selectivity	11-Deoxycortisol, androstenedione, and DHEA produced selectivity indices about 5- to 6-fold higher than cortisol; AVS success increased from about 69% with cortisol to 89% to 91% with alternative steroids	Improves AVS performance but does not replace AVS
Urinary steroidomics	Zhong et al, J Chromatogr B Analyt Technol Biomed Life Sci 2026 ⁷³	37 PA and 104 non-PA patients	Urine; simplified LC-MS/MS	Diagnose PA	Combined urinary steroid profile achieved AUC 0.914, 87% sensitivity, 88% specificity	Small PA cohort; diagnostic rather than robust subtype validation
Urinary steroidomics	Prete et al, J Steroid Biochem Mol Biol 2024 ⁶⁰	158 PA (88 APA, 70 BAH)+65 healthy controls	24-h urine; GC-MS	Diagnose PA; distinguish unilateral vs bilateral disease; infer genotype	Excellent PA vs controls discrimination (AUC, 0.95–0.97); unilateral vs bilateral only modest (AUC, 0.65–0.73); better for <i>KCNJ5</i> -mutated APA (AUC, 0.79–0.85)	Good for detection, weaker for subtype discrimination
Urinary steroidomics	Ulick et al, JCEM 1993 ⁷⁴	42 patients: 21 APA, 21 BAH	Urinary hybrid steroids	Subtype discrimination	Urinary 18-oxocortisol showed marked separation: 2–12 µg/d in BAH vs 17–1203 µg/d in APA	Small cohort; early proof-of-concept
Ang II-based ratios/RAAS peptide profiling	Burrello et al, Hypertension 2020 ⁷⁸	110 hypertensive patients: 33 PA and 77 with hypertension	Serum RAAS Triple-A LC-MS/MS: aldosterone, equilibrium Ang I and equilibrium Ang II	PA screening using AA2R vs renin-based ratios; medication-interference assessment	AA2R AUC comparable with renin-based ratios (AUC range, 0.924–0.970). Optimal cutoffs gave 90% sensitivity and 93% specificity; potential advantage under ACE-inhibitor exposure	Modest single-center cohort; subgroup medication effects require confirmation; no large prospective external validation
Ang II-based ratios/RAAS peptide profiling	Guo et al, Clinical Chemistry 2020 ⁷⁹	78 patients undergoing FST: 56 confirmed PA and 22 non-PA; 312 serial samples	Plasma equilibrium Ang II by LC-MS/MS, compared with PRA and DRC	Analytical validation of equilibrium Ang II and comparison of AA2R with ARR under dynamic RAAS conditions	Equilibrium Ang II was below functional assay sensitivity in 7.4% of samples, vs 43.9% for DRC and 15.1% for PRA. AA2R correlated closely with ARR (≈0.878 with DRC; ≈0.880 with PRA)	Assay-validation study rather than a definitive screening-outcome study; specialized FST cohort; clinical impact on diagnostic accuracy remains to be validated

(Continued)

Table. Continued

Biomarker class	Study	Cohort/design	Matrix/platform	Main clinical task	Main findings/performance	Main limitations
Ang II-based ratios/RAAS peptide profiling	Lo et al, Clin Chem Lab Med 2025 ⁸⁰	Archived plasma samples from 152 hypertensive patients tested for PA	Plasma LC-MS/MS measurement of aldosterone, Ang II, ARR, and AA2R	Comparison of plasma AA2R with ARR for PA screening	AAIIR and ARR showed comparable discrimination: AUC 0.94 for AAIIR and 0.94 for ARR. AA2R sensitivity/specificity 91%/85%, vs 88%/86% for ARR; AA2R correlated with ARR ($r=0.79$)	Retrospective archived-sample design; single-center data; larger prospective validation studies needed
Circulating miRNAs	Decmann et al, Front Endocrinol (Lausanne) 2019 ⁸⁹	123 AVS-characterized PA samples; discovery 30 and validation 93	Plasma miRNAs	Distinguish APA from BAH	Of 50 candidates, only 3 miRNAs validated; each showed only modest performance, with sensitivity and specificity around 60%	Limited reproducibility; modest discriminatory power
Circulating miRNAs	Makhnov et al, Front Endocrinol (Lausanne) 2025 ⁵⁸	Serum miRNA study (non-PA 11, BAH 12, and APA 16)	Serum miRNAs	Diagnose PA; identify unilateral PA	5-miRNA model distinguished hypertension from PA with 100% accuracy; another 5-miRNA model identified unilateral PA with 94% accuracy and AUC 0.946	External application to Decmann data set dropped to 63% accuracy with AUC 0.312; poor reproducibility across cohorts
Circulating miRNAs	Vékony et al, Hypertension 2024 ⁹⁰	Discovery in 18 patients with PA; validation in 108 with PA	miRNA profiling	Distinguish APA from BAH	Best 6-miRNA model achieved AUC 87.1%	Limited standardization; small training cohorts
Proteomics	Carvajal et al, Front Immunol 2021 ⁹²	41 adults: 13 normotensive, 17 hypertension, 11 PA	Serum proteomics	Detect PA	AGP1/ORM1 higher in PA; AUC 0.90, 90% sensitivity, 83% specificity; combining AGP1 with urinary EV miR-21-5p improved AUC to 0.94	Very small PA cohort; subtype work-up not uniform
Proteomics	Tsai et al, Eur J Clin Invest 2025 ⁹³	52 unilateral PA and 46 with hypertension	Plasma proteomics	Distinguish unilateral PA from non-PA; monitor response	6-peptide signature achieved AUC 0.92; validation cohort sensitivity/specificity 81.8%/80.0%; risk scores fell after adrenalectomy	Subtype-specific generalizability limited
Proteomics	Zhou et al, Eur J Endocrinol 2024 ⁵⁷	36 APA, 24 BAH, 30 with hypertension	Plasma targeted multiplex LC-MS/MS	Detect PA; distinguish APA from BAH	5-Protein panel plus renin and aldosterone achieved AUC 0.986 for PA vs hypertension; AUC 0.922 for APA vs BAH with 89% sensitivity and 92% specificity	Modest sample size; independent validation still limited
Proteomics	Makhnov et al, Endocrine 2025	84 tested for PA: 29 hypertension, 29 BAH, 26 APA	Serum proteomics	Three-class discrimination	Ridge-logistic model combining 9 proteins with ARR achieved average AUC 0.991, specificity 97%, sensitivity 95%, accuracy 93%	External validation lacking; circulating proteins may reflect organ damage or medication effects; no true gold standard for bilateral hyperplasia
Multomics	Reel et al, EBioMedicine 2022 ⁹⁷	Multicohort endocrine-hypertension study 307 subjects (PA=113, PPGL=88, CS=41 and hypertension=112)	Plasma miRNAs, plasma steroids, urinary steroids, metanephries, small metabolites	Multiclass endocrine-hypertension classification	Multomics classifiers improved performance by about 11% over best monomics model, reaching about 92% balanced accuracy, 96% specificity, AUC 0.95 for multiclass classification, and AUC 0.96 for endocrine vs primary hypertension	Retrospective, selected cohorts; heterogeneous platforms; limited external validation

AA2R indicates aldosterone-to-Ang II ratio; APA, unilateral primary aldosteronism; AUC, area under the curve; BAH, bilateral primary aldosteronism; CS, Cushing syndrome; FST, fludrocortisone suppression testing; LC-MS/MS, liquid chromatography with tandem mass spectrometry; and PPGL, pheochromocytoma and paraganglioma.

half-life (>60 minutes), together with the influence of cortisol-binding globulin, results in relatively low adrenal-to-peripheral venous gradients and can underestimate diagnostic accuracy. In contrast, adrenal biomarkers with more rapid clearance, such as metanephrine, show a 6-fold higher adrenal-to-peripheral gradient than cortisol and have been shown to be superior for assessment of AVS selectivity.^{66–70} In this direction, in a multicenter cross-sectional study of 229 patients with PA, LC-MS/MS measurement of 11-deoxycortisol, androstenedione,

and DHEA produced selectivity indices about 5- to 6-fold higher than cortisol and increased AVS success rates from about 69% with cortisol to 89% to 91% with the alternative steroids.⁷¹ In particular, 11-deoxycortisol reduced failed AVS procedures by about 54% to 72%, while sensitivity reached 99.8% to 100% versus 97.5% for cortisol, with specificity remaining near 100% and lateralization showing more than 90% concordance across markers, indicating that improved selectivity did not compromise subtype classification.

Urinary Steroids

Urinary steroid profiling has long been recognized as a useful noninvasive approach for characterizing adrenal steroidogenesis (Table). Early GC-MS studies in mixed adrenocortical tumor cohorts showed that unilateral PA was associated with increased urinary metabolites of 18-hydroxycorticosterone and aldosterone, supporting the principle that distinct urinary steroid fingerprints can separate adrenal pathologies.⁷² In a newer simplified LC-MS/MS study, Zhong et al⁷³ analyzed 37 patients with PA and 104 non-PA patients, showing that aldosterone, tetrahydroaldosterone, 18-hydroxycorticosterone, 18-oxocortisol, and 18-hydroxycortisol were all higher in PA, and that the combined urinary steroid profile yielded an AUC of 0.914 with 87% sensitivity and 88% specificity. Prete et al⁶⁰ then applied 24-hour urine steroid metabolomics to 158 patients with PA, including 88 with unilateral and 70 with bilateral PA, together with 65 healthy controls, and showed excellent discrimination of PA from controls with AUC-ROC values of 0.95 to 0.97. However, discrimination of unilateral from bilateral disease was only modest, with AUC-ROC values of 0.65 to 0.73, whereas identification of *KCNJ5*-mutated unilateral PA was better, with AUC-ROC values of 0.79 to 0.85. In this direction, Ulick et al⁷⁴ studied 42 patients, including 21 with unilateral and 21 with bilateral PA, and demonstrated a striking separation in urinary 18-oxocortisol excretion, which ranged from 2 to 12 µg/d in bilateral PA versus 17 to 1203 µg/d in unilateral PA, thereby establishing the potential of urinary hybrid steroids for PA subtype discrimination. Taken together, these data support urinary steroid profiling as a promising noninvasive adjunct for PA detection and selected subtype classification, but not yet a replacement for established routine care diagnostics, as the literature remains limited by small or selected cohorts, heterogeneous assays, and insufficient prospective validation in unselected hypertensive populations.

ANGIOTENSIN PEPTIDOME

Within the angiotensin peptidome, Ang I (angiotensin I) represents the closest biochemical readout of renin activity, although its measurement remains sensitive to preanalytical conditions and postsampling peptide degradation.^{75,76} Ang II (Angiotensin II), the downstream bioactive peptide that directly stimulates aldosterone secretion, provides a more proximal marker of the physiological drive that should regulate aldosterone output.⁷⁷ This rationale underpins the aldosterone-to-Ang II ratio, which captures the inappropriate persistence of aldosterone secretion despite suppressed Ang I and Ang II activity in PA. Burrello et al⁷⁸ first demonstrated the clinical value of this approach using LC-MS/MS-based RAAS Triple-A analysis, reporting 90% sensitivity and

93% specificity for PA screening at an aldosterone-to-Ang II ratio cutoff of 6.6. Guo et al⁷⁹ further supported equilibrium Ang II as an analytically useful marker in suppressed-renin states, while Lo et al⁸⁰ reported comparable diagnostic performance for the endogenous plasma aldosterone-to-Ang II ratio and ARR, with AUCs of 0.938 and 0.935, respectively. In parallel, Demeuse et al⁸¹ validated sensitive LC-MS/MS quantification of Ang 1–7, Ang 1–8, Ang 1–9, and Ang 1–10, extending angiotensin profiling beyond the classical Ang I and Ang II axis toward the ACE2-linked alternative RAAS pathway. Together, these studies support Ang II-based ratios as promising PA screening tools, particularly when renin is analytically suppressed and complicates ARR interpretation. However, prospective validation, standardized preanalytics and assay harmonization remain necessary before routine clinical implementation.

This same conceptual framework is now expanding from diagnostic ratios toward integrated angiotensin-steroid platforms that combine renin activity, angiotensin peptide biology, aldosterone output, and adrenal steroid signatures into a broader multianalyte phenotype. Recent workflows illustrate this transition. Li et al⁸² developed an automated magnetic bead-assisted LC-MS/MS platform for angiotensin peptides, aldosterone, and 18-hydroxylated steroid derivatives, whereas Wang et al⁸³ improved ultrasensitive enrichment and detection of Ang I, Ang II, Ang III, and aldosterone, although without broader adrenal steroid profiling. Collectively, these platforms may support not only PA diagnosis and subtype stratification, but also precision therapeutics. Future value of the technology may lie in refining surgical versus medical treatment selection, monitoring mineralocorticoid receptor antagonist response, and assessing biochemical remission after adrenalectomy. At present, however, robust reference intervals, harmonized cutoffs, standardized preanalytical protocols, and prospective outcome-based validation are still required before these approaches can be adopted in routine clinical care.

MICRORNAs

miRNAs have emerged as attractive candidate biomarkers in PA (Figure 2), as they are relatively stable in blood and may capture upstream regulatory disturbances that are not fully reflected by aldosterone or steroid measurements alone.⁸⁴ Experimental studies have shown that Dicer-dependent miRNA processing constrains adrenal steroidogenesis, that miR-24 directly modulates *CYP11B2* expression and aldosterone production,⁶² and that several miRNAs, including miR-125a-5p and miR-125b-5p, regulate corticosteroidogenic gene expression.⁸⁵ In addition, miR-21 is induced by Ang II and promotes both aldosterone secretion and adrenal cell proliferation.⁸⁶ Tissue-based studies further support the role of miRNA dysregulation

in aldosterone-producing lesions. For example, unilateral PA has been associated with lower expression of miR-24-5p and miR-10b-5p. Reduced miR-375 has also been reported and may exert tumor-suppressive effects through metadherin. Moreover, different miRNA expression patterns have been identified among functional adrenocortical adenomas⁸⁷ and between functional and nonfunctional adrenocortical adenomas.⁸⁸ However, the aforementioned data remain largely mechanistic and exploratory. Most studies are small, often rely on microarray-era platforms, use heterogeneous comparator tissues, and are enriched for surgically treated unilateral disease, limiting their direct applicability to routine diagnosis or to the broader clinical spectrum of PA.

Clinical studies of circulating miRNAs have yielded mixed results (Table). In the landmark study by Decmann et al,⁸⁹ AVS-characterized PA plasma samples were analyzed in 123 patients, including a 30-sample discovery cohort (16 unilateral, 14 bilateral PA) and an independent 93-sample validation cohort. Although 50 candidate miRNAs were identified initially, only 3 (miR-30e-5p, miR-30d-5p, and miR-7-5p) were validated, all showing higher expression in bilateral than unilateral PA. However, the diagnostic performance of each individual miRNA was only modest, with sensitivity and specificity around 60%. More recent studies have reported considerably stronger internal performance. In particular, in a cohort of 11 non-PA, 12 with bilateral and 16 with unilateral PA, Makhnov et al⁵⁸ identified 33 significantly dysregulated serum miRNAs and derived a 5-miRNA model that distinguished hypertension from PA with 100% accuracy, and another 5-miRNA model that identified unilateral PA with 94% accuracy and an AUC of 0.946. However, external application to the Decmann data set yielded only 63% accuracy with an AUC of 0.312, and qPCR confirmed only 6 overlapping miRNAs from the sequencing analysis. Likewise, Vékony et al⁹⁰ conducted microRNA profiling to differentiate unilateral from bilateral disease, in 18 discovery samples from patients with PA. A best 6-mRNA model reported an AUC of 87.1% for subtype classification in a validation cohort of 108 patients. Okamoto et al⁹¹ further identified miR-222-3p as significantly higher in unilateral than bilateral PA in both adrenal venous blood and the inferior vena cava in a limited number of included cases, with positive correlations of miR-222-3P with adrenal cell proliferation, tumor diameter, and aldosterone secretion, but this remained an exploratory single-marker study without a clinically validated cutoff. Overall, the literature suggests that miRNAs play a biologically plausible role and may potentially serve as informative complementary biomarkers for PA detection and lateralization, yet their clinical translation is still constrained by poor reproducibility across cohorts, differences between plasma and serum, preanalytical variability, normalization

and platform effects, and the lack of large, standardized prospective validation.

PROTEOMICS

Circulating proteomics in PA has moved from proof-of-concept screening markers toward subtype-oriented plasma signatures (Figure 2). In an early cross-sectional study of 41 adults (13 normotensive controls, 17 with primary hypertension, and 11 with PA), Carvajal et al⁹² showed that serum alpha-1-acid glycoprotein 1 (AGP1/ORM1) was markedly higher in PA than in the comparator group without disease (median, 934.1 versus 62.5 and 60.7 µg/mL), with an AUC of 0.90, 90% sensitivity, and 83% specificity for identifying PA. Importantly, combining AGP1 with urinary extracellular-vesicle miR-21-5p improved the AUC to 0.94 with 90% sensitivity and 85% specificity (Table). The main value of this study was to show that blood proteins can capture aldosterone-related systemic biology beyond renin and aldosterone alone, but its diagnostic implications remained preliminary because only 11 PA cases were included and subtype work-up was not uniform. More recently, Tsai et al⁹³ extended this concept using deep plasma proteomics in 52 patients with unilateral PA and 46 with essential hypertension, identifying a 6-peptide signature (HBB, FIBA, CO7, ALBU, C4BPA, and A2AP) that achieved an AUC of 0.92, with sensitivity and specificity of 80.5% and 83.3%, respectively, in the training cohort and 81.8% and 80.0% in the validation cohort. Importantly, risk scores fell after adrenalectomy, which supports disease specificity and suggests potential value for treatment monitoring.

The strongest proteomic data for subtyping come from Zhou et al,⁵⁷ who developed a targeted multiplex LC-MS/MS workflow after depleting abundant plasma proteins and quantified >1500 proteins per sample (Table). In their final clinically curated cohort of 36 AVS/surgery-defined unilateral PA, 24 bilateral cases, and 30 primary-hypertension controls, 12 candidate proteins showed ROC AUC values >0.7, and a 5-protein panel (CHGA, APOC3, CD56/NCAM1, KRT5, and AZGP1) combined with renin and aldosterone achieved an AUC of 0.922 for unilateral *versus* bilateral PA, with 89% sensitivity and 92% specificity, while also separating PA from primary hypertension with an AUC of 0.986.⁵⁷ This study is particularly important because it linked high-dimensional plasma proteomics to clinically relevant lateralized disease, although independent validation was still limited by the modest sample size. A complementary serum study by Makhnov et al⁵⁸ analyzed 88 hypertensive patients, with 84 evaluable samples after quality control (29 with hypertension, 29 bilateral PA, and 26 unilateral PA), and found 56 differentially expressed proteins overall and 42 between unilateral and bilateral PA. In that data set, F9 best differentiated PA from hypertension, DPP4 best

separated unilateral PA, and HSPB1 best separated bilateral from unilateral disease, while a ridge-logistic model combining 9 proteins with the ARR achieved an average AUC of 0.991, specificity of 98%, sensitivity of 95%, and accuracy of 93%. The major strengths of this study were the 3-class design and orthogonal ELISA confirmation for selected proteins, but the results still require external validation and may partly reflect target-organ damage or medication effects rather than adrenal lesion biology itself. In addition, interpretation of these findings remains limited by the absence of a true reference standard for the diagnosis of bilateral disease, a limitation that applies to all studies in this field.

MULTIOMICS

Multimic studies increasingly suggest that PA is a heterogeneous continuum of aldosterone dysregulation, ranging from subclinical abnormalities to overt disease. By integrating complementary molecular layers beyond genotype alone, multiomics may help refine disease probability, severity assessment, subtype classification, and clinically actionable decision thresholds. At the tissue level, the first deep quantitative proteomic and phosphoproteomic analysis of 6 matched unilateral PA/adjacent cortex pairs showed not only increased abundance of steroidogenic enzymes such as aldosterone synthase, 3β -hydroxysteroid dehydrogenase, and steroid 21-hydroxylase, but also pathway-level dysregulation involving extracellular matrix remodeling, N-glycosylation, GABAergic signaling, RHOC, and mTORC1, indicating that posttranslational control and signaling-network rewiring contribute materially to unilateral PA biology beyond known somatic PVs.⁹⁴ Extracellular vesicle (EV) studies have also provided an additional systems-biology layer linked to vascular injury. In particular, a study of 12 patients with PA, 12 with primary hypertension, and 8 normotensive controls found a 2.2-fold higher EV concentration in PA, more endothelial-derived EVs, and 15 upregulated plus 4 downregulated EV microRNAs, with *EDN1* expressed only in PA.⁹⁵ A later study in 32 patients with PA, 29 with primary hypertension, and 22 controls identified 19 differentially expressed EV surface antigens and showed that the EV signature shifted toward primary hypertension after unilateral adrenalectomy. It also demonstrated that PA-derived EVs induced endothelial expression of *AKT1*, *CALR*, *CSNK2A1*, *FN1*, and *PIK3R1*, supporting a possible mechanistic link with endothelial dysfunction and accelerated organ damage rather than biomarker status alone.⁹⁶ The clearest evidence that integration adds value comes from the multiomics ML study by Reel et al⁹⁷ (Table), which measured 409 features spanning plasma miRNAs, plasma steroids, urinary steroids, metanephrines, and small metabolites across endocrine-hypertension cohorts and showed that multiomics classifiers improved performance by about

11% over the best monomics model, achieving about 92% balanced accuracy, 96% specificity, and an AUC of 0.95 for multiclass endocrine-hypertension classification, and an AUC of 0.96 for endocrine versus primary hypertension. Taken together, these studies support the concept that integrated omics can uncover pathway-level dysregulation and clinically relevant patterns that single-modality testing misses, but translation into routine care is still constrained by small and highly selected cohorts, retrospective design, heterogeneous platforms and pre-processing pipelines, inconsistent subtype reference standards, limited external validation, and uncertainty about whether circulating signals reflect adrenal biology directly or downstream cardiovascular and renal consequences of aldosterone excess.

FUTURE PERSPECTIVES AND CHALLENGES

The updated Endocrine Society guideline broadens screening while keeping aldosterone and renin at the center of PA diagnostics, reflecting a clinically pragmatic approach: these markers provide actionable thresholds for identifying patients who may benefit from further evaluation or PA-specific treatment, even though they do not fully capture the biological continuum, molecular heterogeneity, or subtype complexity of aldosterone dysregulation.¹² This gap matters because PA is increasingly understood as a multidimensional syndrome that spans a continuum from subclinical renin-independent aldosteronism to overt unilateral or bilateral disease. Indeed, gray areas of current diagnostic tests remain, particularly with borderline or atypical biochemical profiles. Novel biomarkers address some of these challenges and define these gray areas more precisely, by capturing biology that current routine care diagnostic tools do not. Steroidomics is currently the most developed of the new approaches because it measures the downstream biochemical output of adrenal steroidogenesis and can identify hybrid steroid patterns linked to genotype and lateralization. Proteomics adds information on inflammatory, vascular, and metabolic consequences of aldosterone excess, while miRNA and EV platforms may capture upstream regulation and intercellular signaling. Their key strength is that they align more naturally with the continuum hypothesis. Indeed, instead of asking whether a patient is simply positive or negative, they generate multidimensional fingerprints with potential to estimate graded probability, molecular subtype, and sometimes even treatment responsiveness. The problem is that most data sets remain retrospective, modest in size, and highly selected, with limited external validation and substantial platform heterogeneity. Analytical pipelines also vary across studies, especially for proteomics and miRNA normalization, which makes reproducibility and transportability difficult.

ML is therefore likely to be more important than individual biomarkers. In PA, single features are not robust enough on their own because the disease is biologically heterogeneous and laboratory measurements are noisy. By contrast, ML models can integrate steroid panels, proteins, miRNAs, imaging features, potassium, blood pressure, and clinical history into probabilistic outputs that better reflect real disease architecture. The major challenge, though, is the robustness of ML models. A recent laboratory medicine analysis of steroidomics-based ML models showed that reproducibility and interlaboratory transport are critical requirements if ML outputs are to be trusted clinically.⁹⁸ More broadly, current ML reviews in PA emphasize the same concerns: black-box behavior, limited interpretability, center-specific training data, and insufficient real-world validation. Future ML models will need to be explainable, and prospectively tested across populations and laboratory platforms. Translation into routine care will depend on infrastructure as much as on biology. Biomarkers and ML models are most likely to succeed when embedded into clinical decision support systems that can automatically ingest assay results, account for sex, age, and assay platform, and return transparent, patient-specific diagnostic probabilities with narrative interpretations for further patient management.⁹⁹ That is, how complex omics data become clinically usable rather than remaining research outputs. But these systems will only be credible if they are built on standardized protocols for sampling, processing, calibration, normalization, and reporting.

What the field now needs is not another wave of small discovery studies, but rigorous implementation science grounded in clinically meaningful outcomes. Large multicenter randomized prospective trials are needed to test whether biomarker-guided and ML-supported strategies improve not just diagnostic accuracy, but also patient outcomes, referral efficiency, surgical selection, cardiovascular risk reduction, and cost-effectiveness. Regulatory acceptance of novel tools will require reproducibility, transparency, and pre-defined performance standards. Until that ecosystem exists, the sensible priority is dual: continue developing integrated biomarkers that better reflect the continuum nature of PA, while simultaneously making present-day routine care diagnostic tools more accurate, more standardized, and less vulnerable to assay-driven misclassification.

ARTICLE INFORMATION

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